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The one-stage design for nationwide studies on temperature and health

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Abstract

Nationwide studies investigating the short-term health effects of ambient temperature commonly use a two-stage design that estimates location-specific associations before combining them through a meta-analysis. Although flexible, this approach can be statistically complex and computationally intensive, particularly when dealing with multiple locations or sparse data. We present a simplified one-stage design modeling the temperature–mortality association directly at the national level by aggregating data across locations and using a country-averaged exposure metric. We applied and compared the one-stage and two-stage designs using daily all-cause mortality and ambient temperature data from England and Wales, and from Spain (1993–2006). The one-stage design produced consistent national estimates with fewer modeling steps and reduced computational requirements. While it offers practical advantages, especially in settings with limited location-level data or analytical capacity, it also assumes a homogenous temperature-mortality association and does not allow to explore spatial heterogeneity. Overall, the one-stage design is a useful tool for national-level assessments of temperature-related health risks, providing a more accessible, and timely alternative to complex multi-location models when geographical variation is not the primary focus.

1. Background

Assessing the short-term health effects of environmental exposures, such as ambient temperature, often requires nationwide epidemiological studies that account for geographical and temporal variations. Traditionally, a two-stage design has been widely used, where location-specific associations are first estimated and then combined in a second-stage meta-analysis [1]. While the two-stage design allows for flexibility in capturing geographical heterogeneity, it also introduces statistical complexity requiring careful methodological choices [2, 3], and large sample sizes in each location to produce stable estimates [4].

An alternative is a simplified one-stage design that models the exposure-response association directly at the national level using a country-aggregated outcome and a country-averaged exposure metric. This approach provides a more direct framework for national assessments, particularly when the primary objective is to quantify an overall nationwide association rather than to characterize geographical variability. Despite its application in several national temperature-mortality studies [5–10], the one-stage design has not yet been formally described or conceptualized. In this paper, we aim to articulate its conceptual basis, discussing key assumptions and limitations, and illustrate its application in comparison with the two-stage approach.

2. Example data sets

As illustrative examples, we used two datasets that have been widely analyzed in previous temperature–mortality studies, the 10 regions of England and Wales and the 52 provincial capital cities in Spain, both for the study period 1993–2006 [11–14]. In England and Wales, temperature exposure is relatively homogeneous (supplementary figure S1). The average daily mean temperature across regions was 10.4 °C, ranging between 9.5 °C and 11.7 °C. The mean daily mortality count was 148.1 deaths, ranging between 79.8 and 221.5 (supplementary table S1). In contrast, temperature exposure in Spain is more heterogeneous across cities (supplementary figure S2). The average daily mean temperature was 15.5 °C, ranging between 11.1 °C and 21.7 °C, and the mean daily mortality was 6.3 deaths, ranging between 0.8 and 69.2 (supplementary table S2).

3. The one stage design

The one-stage design simplifies the estimation of short-term health effects of environmental exposures (e.g. ambient temperature, air pollution) at the nation-level by directly linking exposure and health outcomes in a single regression, rather than modeling locations separately and synthesizing results through meta-analysis as in the two-stage approach. The one-stage design can be expressed as:

$$Y_t = \alpha + f(X_t, \theta) + s(t, \lambda) + Z_t\gamma + \epsilon_t$$

where: Y_t is the nationwide aggregated daily health outcome count at time t (e.g. sum of daily mortality counts across all locations), X_t is the nationwide averaged exposure (e.g. the average daily ambient temperature across all locations) modeled using a flexible function $f(X_t, \theta)$ (e.g. a spline or distributed lag non-linear model (DLNM)) to capture potential nonlinear and lagged effects, Z_t is a vector of additional confounders (e.g. weekdays, public holidays) with coefficients γ , and ϵ_t is the residual error term.

The application of the one-stage design relies on several key assumptions:

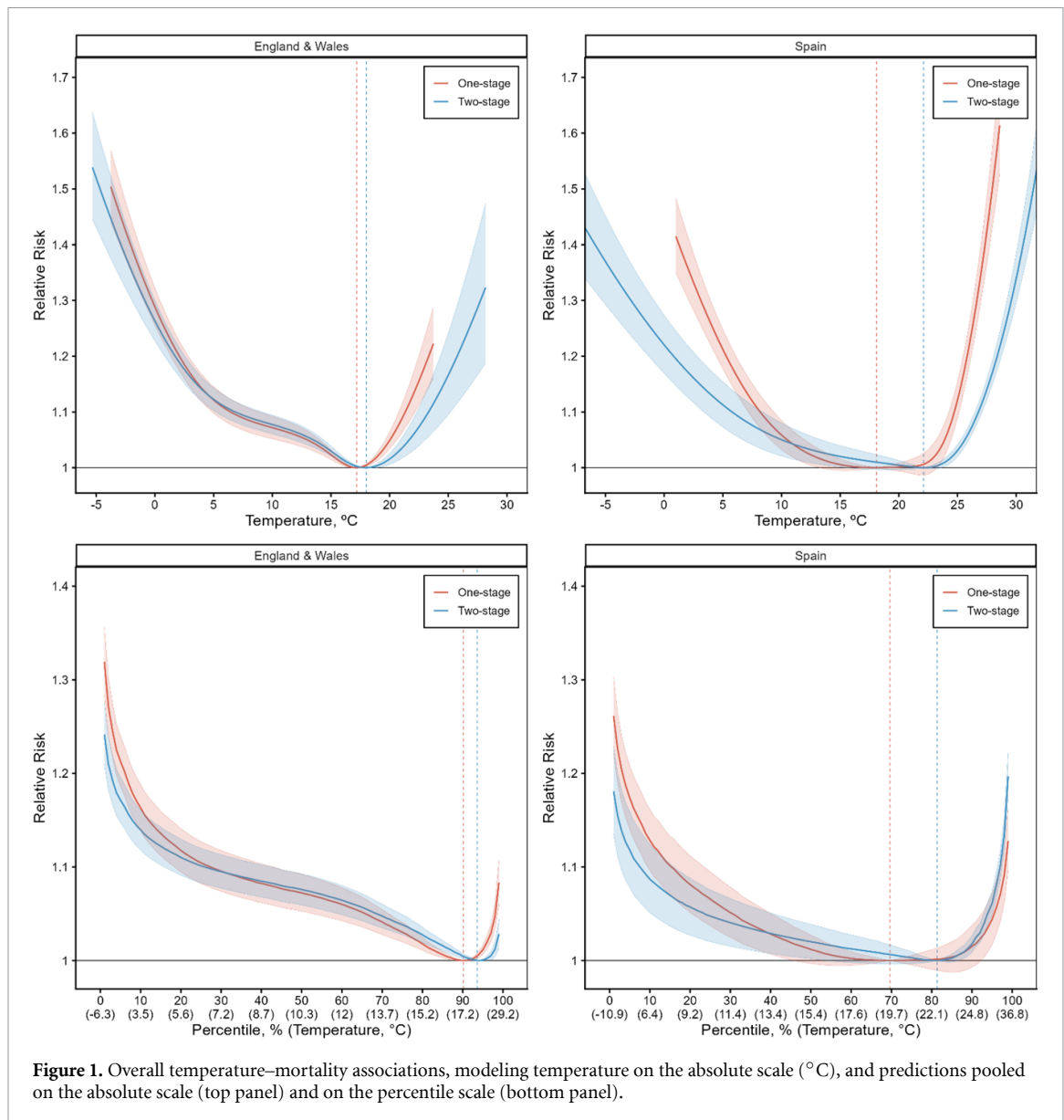
- (a) A biologically plausible exposure-response relationship at the national-level, which constitutes the main target of inference. This condition holds as long as there is no evidence suggesting the presence of incompatible exposure-response patterns within the country.
- (b) Adequate representation of exposure. The aggregated exposure metric accurately reflects the national distribution of exposure, ensuring that the exposure variable captures the relevant variability for estimating the national association and avoiding bias in the estimated exposure-response curve.
- (c) Sufficient aggregation of outcome data. The aggregated health outcome reliably reflects the underlying population-level effect. Variability in data quality, reporting practices, and extreme local values should not distort the national estimate. Proper aggregation is required for the one-stage model to estimate an unbiased national-level association.

4. Comparison of one-stage and two-stage designs

4.1. Modeling temperature on the absolute scale

In the one-stage design, we initially fitted a quasi-Poisson regression model to the nationwide daily death counts, modeling averaged ambient temperature on the absolute scale (in °C). To control for long-term trends and seasonality, the model included a natural cubic spline of time with 10° of freedom per year, along with indicator variables for day of the week. The temperature-mortality association was modeled using a DLNM. Specifically, the exposure-response relationship was captured with a natural cubic spline, placing internal knots at the 10th, 75th, and 90th percentiles of the temperature distribution. The lag-response relationship was also modeled using a natural cubic spline, with three internal knots placed at equally spaced values on the log scale. The lag period was extended to 21 d to account for the delayed effects of cold. These modeling choices follow previously published analyses of these datasets [11–14].

For the two-stage design, the same modeling choices were applied to each location in the first stage, while in the second stage, location-specific estimates were pooled to represent the overall exposure-response relationship by cumulating risks across the lag period for each country. A random-effects multilevel meta-analytical model was employed to account for between-location heterogeneity in risk estimates. To ensure full comparability with the one-stage formulation, we did not include any meta-regressors in the second stage. Likewise, for the same reason, we adopted maximum likelihood rather



than restricted maximum likelihood for the second-stage estimation [3]. Based on the pooled estimates obtained from the multivariate meta-regression, we reconstructed the temperature-mortality exposure-response relationship. To this end, we defined a common absolute temperature scale to reflect the countrywide temperature distribution based on the location-specific temperature distributions and generated the corresponding basis matrix using the same spline functions specified in the first-stage models.

Figure 1 (top panel) compares the temperature-mortality associations estimated using the one-stage and two-stage designs. In England & Wales, both designs produced similar exposure-response curves; however, the one-stage model estimated a slightly lower minimum mortality temperature (MMT) and a steeper increase in mortality risk, particularly in response to heat. In Spain, the exposure-response curves differed more substantially, showing a noticeable shift in the position of the MMT and in the magnitude of relative risks, with the one-stage model estimating substantially higher mortality risks across much of the temperature range.

The use of an absolute temperature scale limits the comparability of resulting curves across modeling approaches, as the one-stage design relies on a narrower, nation-level average exposure distribution. To enhance comparability, we reconstructed the temperature-mortality exposure-response relationships using a common percentile scale based on the location-specific temperature distributions. Specifically, the results are expressed in terms of nationwide temperature percentiles, which correspond to different absolute temperatures across cities. This approach, commonly adopted in multi-location studies [15], enables the pooling of city-specific associations while accounting for local climatic differences.

Figure 1 (bottom panel) shows overall exposure-response curves plotted on a percentile scale, allowing direct comparison between designs. In England and Wales, both designs produce close MMTs and cold-related risk estimates; however, the one-stage design shows higher heat-related risks. In Spain, the curves from both designs are now more aligned in shape and pattern, but notable differences remain in the MMT.

4.2. Temperature modeled on the percentile scale

When comparing temperature-mortality relationships across cities, especially those with different climates, using absolute temperature values can be problematic. This issue is particularly relevant in the two-stage design, where temperature ranges often do not overlap, such as in Spanish cities. This issue arises because the national-level temperature-mortality curve is reconstructed using a common basis function, which must be compatible with the basis functions used for each city's individual curve. When temperature distributions differ substantially, aligning these functions becomes challenging.

To address this constraint, we reparametrized both the one-stage and two-stage regression models using percentiles derived from the temperature distributions, rather than absolute temperature values. In the one-stage design, the association was modeled using the percentile distribution of the nationwide averaged ambient temperature. In the two-stage design, location-specific associations were similarly re-estimated using each location temperature percentiles in the first stage, with the resulting coefficients pooled in the second stage using the same multivariate meta-analytical framework. This transformation allowed for standardized comparison of temperature effects across locations by aligning relative positions within each city's temperature distribution, rather than relying on shared absolute values. It also supported a more interpretable and consistent pooling of results in multi-location studies, where the same percentile reflects comparable relative heat or cold stress, even if the corresponding absolute temperature differs across cities.

Figure 2 illustrates the improvement in comparability between the one-stage and two-stage designs after reparametrizing temperatures on the percentile scale. In England & Wales, both models produced nearly identical exposure-response curves and matched MMTs, indicating full alignment between the two approaches. In Spain, the agreement between the shapes of the curves also improved, with much closer estimates of the MMT and mortality risks associated with non-optimal temperatures. However, the two-stage approach estimates slightly higher risks from moderate cold temperatures compared to the one-stage model.

5. Discussion

This study introduces the one-stage modeling approach as a simplified alternative to the traditional two-stage design for assessing short-term temperature-related health effects at the national level. By relying on aggregated health outcome data and an averaged exposure metric, it enables estimation of a single nationwide association without the need for location-specific models, thereby reducing analytical complexity and improving accessibility for public health analysis.

5.1. Applicability, strengths and advantages

The one-stage design is appropriate when the primary objective and the available aggregated data structure are limited to estimating an overall national-level association. This assumes a biologically plausible common association at the country level and that the average exposure metric adequately captures national exposure patterns, with no scope to model between-location heterogeneity or implement second-stage extensions typical of the two-stage framework.

The example from England and Wales shows that when conditions are approximately homogeneous across regions, in terms of both exposure patterns and exposure-outcome associations (figure S3), one-stage and two-stage approaches produce nearly identical national summaries. Moreover, in Spain this variability is not fully explained by exposure heterogeneity alone, as city specific exposure-outcome curves differ in shape (figure S4), consistent with heterogeneity in the estimated associations. As a result, national estimates differ slightly but remain broadly comparable. This aligns with the individual participant data meta-analysis literature, where one-stage and two-stage procedures are mathematically equivalent when the same model is applied across units. Riley *et al* [16] showed that, under a fixed-effects model, the one-stage and two-stage summary estimates are identical, while Mathew and Nordström [17] extended this equivalence to settings with more general covariance structures.

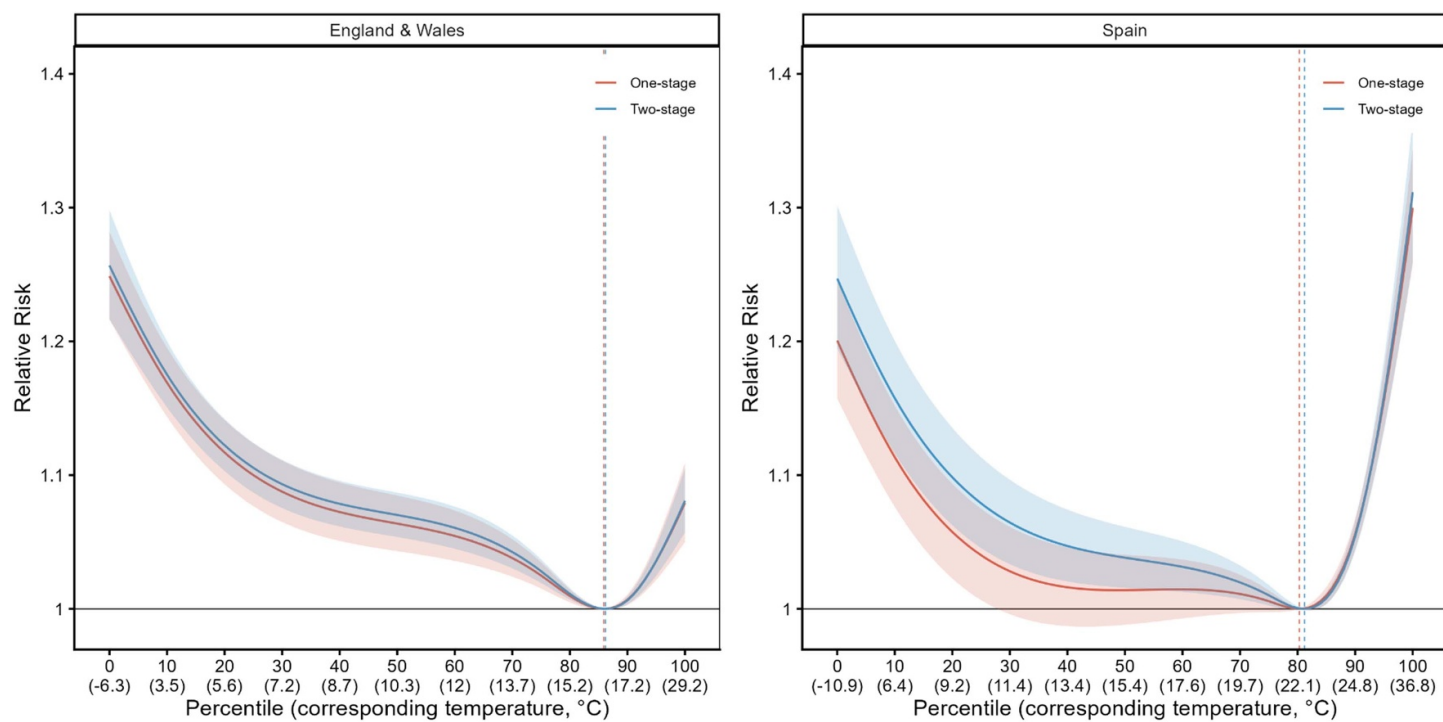


Figure 2. Overall temperature-mortality associations, modeling temperature on the percentile scale.

Although the one-stage model uses aggregated rather than individual data, when the exposure–response relationship is homogeneous across units the aggregated exposure functions act as sufficient predictors. Under these conditions, the one-stage formulation becomes mathematically analogous to a simplified individual-level one-stage framework that maximizes the same likelihood as the fixed-effect two-stage estimator [18]. Consequently, discrepancies between the one-stage and two-stage results are most likely driven by differences in modeling choices rather than by intrinsic distinctions between the two designs [18].

When the exposure–response relationship is reasonably homogeneous and the aggregated exposure reflects the national distribution, the one-stage approach becomes particularly advantageous. This applies notably to climatically homogeneous countries with many locations but sparse outcome data, where a two-stage design would involve fitting numerous local models and the pooled association could be influenced by unstable estimates from small locations. In addition, when location-specific exposure data are available, the one-stage design can be refined using a population-weighted average, which more accurately reflects national exposure and is now often feasible through high-resolution reanalysis datasets such as ERA5 [19, 20]. This weighting reduces the risk of misrepresenting sparsely populated or poorly monitored areas, thereby enhancing the validity and applicability of the national estimate.

There are settings in which the one-stage design is the only analytical option. This occurs when health outcome data are available only in aggregated form because location-specific counts cannot be accessed due to privacy, administrative restrictions, or reporting limitations. In such cases, the one-stage design provides a practical and methodologically sound way to estimate a national-level exposure–response association directly from the available aggregated time-series data.

5.2. Limitations and considerations

A key limitation of the one-stage approach arises when either the temperature–mortality relationship or the exposure itself varies substantially across locations. This situation is common in countries with diverse climates, populations, or vulnerability profiles. In such cases, the one-stage framework may obscure important local differences and misrepresent the true effects in individual locations, reflecting the ecological fallacy in which associations observed at the national level do not necessarily represent individual- or location-specific relationships [21]. Moreover, because the model aggregates data across locations without accounting for between-location variability, the confidence intervals for the national estimate may be narrower than they would be if heterogeneity were incorporated, potentially resulting in overstated precision [22].

The one-stage design also does not allow for the investigation of heterogeneity between locations, and a single national estimate cannot capture variations in risk driven by context-specific modifiers such as urbanization, adaptation, or healthcare access. In such settings, a two-stage approach is preferable because it enables the investigation of sources of heterogeneity, the estimation of location-specific effects, and the calculation of local attributable risks for health impact assessments, thereby better representing the complexity of temperature-related health risks.

However, it is important to recognize that applying a two-stage approach in the presence of heterogeneity does not guarantee that this heterogeneity is properly addressed. The validity of the pooled association depends on several modeling choices, many of which are subtle and often overlooked. For example, two-stage analyses typically assume the same seasonal adjustment across all locations, which may inadequately control temporal patterns when local seasonality differs. Location-specific estimates may also be misaligned when using an absolute temperature scale or identical degrees of freedom and estimates from smaller locations may be unstable yet still receive substantial weight, potentially masking true variability. In addition, the pooled exposure–response curve reflects both the pooled coefficients and the chosen spline basis, which may differ from those used in the local models and substantially affect the shape of the reconstructed curve. Finally, even when notable heterogeneity is detected, the pooled curve may still be biased if it is interpreted as representing all locations equally.

An alternative to the one-stage and two-stage designs is provided by a high-resolution spatiotemporal one-stage modeling framework that uses fine-scale data to enable the simultaneous estimation of temperature–mortality associations across space and time within a unified structure [23, 24]. This approach can capture local heterogeneity more accurately than national aggregated models and may address some of the limitations of two-stage meta-analyses. However, they require high-resolution exposure and health outcome data and are often computationally demanding, which may limit their practical use in many settings.

6. Conclusion

The one-stage design provides a practical and efficient framework for assessing the short-term health effects of ambient temperature at the national level. Its ability to pool data across all regions and generate stable national estimates is particularly valuable for small countries with limited regional information. Although it may not capture regional heterogeneity, its computational efficiency, conceptual simplicity, and broad applicability make it a compelling option for nationwide analyses.

Data availability statement

The dataset for England and Wales is available as open access in the GitHub repository: https://github.com/gasparrini/2015_gasparrini_Lancet_Rcodedata/. The dataset for Spain is not publicly available due to restrictions from the Spanish National Statistics (INE).

The data that support the findings of this study are openly available at the following URL/DOI: <https://github.com/aureliotobias/onestage/> [25].

Supplementary Data available at <https://doi.org/10.1088/2752-5309/ae57c9/data1>.

Author contributions

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Conceptualization (equal), Data curation (equal), Formal analysis (equal), Methodology (equal), Software (equal), Writing – original draft (equal)

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Conceptualization (equal), Data curation (equal), Formal analysis (equal), Methodology (equal), Software (equal), Writing – review & editing (equal)

References

- [1] Sera F and Gasparrini A 2022 Extended two-stage designs for environmental research *Environ. Health* **21** 41
- [2] Gasparrini A and Armstrong B 2013 Reducing and meta-analysing estimates from distributed lag non-linear models *BMC Med. Res. Methodol.* **13** 1
- [3] Sera F, Armstrong B, Blangiardo M and Gasparrini A 2019 An extended mixed-effects framework for meta-analysis *Stat. Med.* **38** 5429–44
- [4] Armstrong B G, Gasparrini A, Tobias A and Sera F 2020 Sample size issues in time series regressions of counts on environmental exposures *BMC Med. Res. Methodol.* **20** 15
- [5] Kunst A E, Looman C W N and Mackenbach J P 1993 Outdoor air temperature and mortality in The Netherlands: a time-series analysis *Am. J. Epidemiol.* **137** 331–41
- [6] Lubczyńska M J, Christophi C A and Lelieveld J 2015 Heat-related cardiovascular mortality risk in Cyprus: a case-crossover study using a distributed lag non-linear model *Environ. Health* **14** 39
- [7] Alahmad B, Shakarchi A, Alseaidan M and Fox M 2019 The effects of temperature on short-term mortality risk in Kuwait: a time-series analysis *Environ. Res.* **171** 278–84
- [8] Folkerts M A, Bröde P, Botzen W J W, Martinius M L, Gerrett N, Harmsen C N and Daanen H A M 2020 Long term adaptation to heat stress: shifts in the minimum mortality temperature in the Netherlands *Front. Physiol.* **11** 225
- [9] Jahan S, Cauchi J P, Galdies C, England K and Wraith D 2022 The adverse effect of ambient temperature on respiratory deaths in a high population density area: the case of Malta *Respir. Res.* **23** 299
- [10] Ordanovich D, Tobias A and Ramiro D 2023 Temporal variation of the temperature–mortality association in Spain: a nationwide analysis *Environ. Health* **22** 5
- [11] Armstrong B G, Chalabi Z, Fenn B, Hajat S, Kovats S, Milojevic A and Wilkinson P 2011 Association of mortality with high temperatures in a temperate climate: England and Wales *J. Epidemiol. Community Health* **65** 340–5
- [12] Gasparrini A, Armstrong B, Kovats S and Wilkinson P 2012 The effect of high temperatures on cause-specific mortality in England and Wales *Occup. Environ. Med.* **69** 56–61
- [13] Tobias A, Armstrong B and Gasparrini A 2017 Investigating uncertainty in the minimum mortality temperature: methods and application to 52 Spanish cities *Epidemiology* **28** 72–76
- [14] Roye D, Iniguez C and Tobias A 2020 Comparison of temperature-mortality associations using observed weather station and reanalysis data in 52 Spanish cities *Environ. Res.* **183** 109237
- [15] Gasparrini A *et al* 2015 Mortality risk attributable to high and low ambient temperature: a multi-country observational study *Lancet* **386** 369–75
- [16] Riley R D, Lambert P C and Abo-Zaid G 2010 Meta-analysis of individual participant data: rationale, conduct, and reporting *BMJ* **340** c221
- [17] Mathew T and Nordström K 2010 Comparison of one-step and two-step meta-analysis models using individual patient data *Biom. J.* **52** 271–87
- [18] Burke D L, Ensor J and Riley R D 2017 Meta-analysis using individual participant data: one-stage and two-stage approaches, and why they may differ *Stat. Med.* **36** 855–75

- [19] Kennard H, Oreszczyn T, Mistry M and Hamilton I 2022 Population-weighted degree-days: the global shift between heating and cooling *Energy Build.* **271** 112315
- [20] de Schrijver E, Folly C L, Schneider R, Royé D, Franco O H, Gasparrini A and Vicedo-Cabrera A M 2021 A comparative analysis of the temperature-mortality risks using different weather datasets across heterogeneous regions *Geohealth* **5** e2020GH000363
- [21] Robinson W S 1950 Ecological correlations and the behaviour of individuals *Am. Sociol. Rev.* **15** 351–7
- [22] Gasparrini A, Armstrong B and Kenward M G 2012 Multivariate meta-analysis for non-linear and other multi-parameter associations *Stat. Med.* **31** 3821–39
- [23] Quijal-Zamorano M, Martinez-Beneito M A, Ballester J and Mari-Dell’Olmo M 2024 Spatial Bayesian distributed lag non-linear models (SB-DLNM) for small-area exposure-lag-response epidemiological modelling *Int. J. Epidemiol.* **53** dyae061
- [24] Barceló M A and Sáez M 2026 Assessing excess mortality and heat-attributable risk during the summer of 2022 in Catalonia, Spain: a Bayesian spatiotemporal analysis *J. Geogr. Syst.* **28** 123–49
- [25] Tobías A and Íñiguez C 2026 One-stage design [R scripts] *GitHub* (available at: <https://github.com/aureliotobias/onestage/>)